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$$f(x, y) = 4xy - 2x^4 - y^2$$

$$\frac{\partial f}{\partial x} = 4y - 8x^3 = 0$$

$$\Leftrightarrow x = 8x(1 - x^2) = 0 \Leftrightarrow 8x(1 - x)(1 + x)$$

$$\frac{\partial f}{\partial y} = 4x - 2y = 0$$

$$\Rightarrow y = 2x$$

$$\text{Dus: } x = 0 \Rightarrow y = 0, x = 1 \Rightarrow y = 2, x = -1 \Rightarrow y = -2,$$

$$\text{Punten: } (0, 0), (1, 2), (-1, -2)$$

$$H(x, y) = \begin{vmatrix} -24x^2 & 4 \\ 4 & -2 \end{vmatrix} \Rightarrow 48x^2 - 16$$

$$H(0, 0) = -16 \Rightarrow \text{zadelpunt}$$

$$H(1, 2) = 32 \Rightarrow \frac{\partial^2 f}{\partial x^2}(1, 2) = -24 < 0 \Rightarrow \text{Maximum}$$

$$H(-1, -2) = 32 \Rightarrow \frac{\partial^2 f}{\partial x^2}(-1, -2) = -24 < 0 \Rightarrow \text{Maximum}$$

References